(1.) Course Title: INFPHD641 Developing web applications with maps (complex exam preparatory)	**Credits: 6**
Course **Classification: Mandatory**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: ….. (credit%)	
Type of Class: practice and **number of hours**: **2** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: …**English**……)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: ……..	
Assessment Method (Exam / Midterm Grade / **Other**[3]): …….. **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: ……..	
Course Placement in Curriculum (Semester Number): …	
Prerequisites (*if any*): ……..	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

The exam focuses on the principles, technologies, and applications of web mapping APIs and geospatial services. Students are expected to demonstrate knowledge of the most commonly used open-source web mapping APIs, including their advantages and limitations. Topics include raster map data formats and sources, such as TMS, WMS, and georeferenced images, as well as vector geodata formats commonly used on the client side, including GeoJSON and vector tiles. Students should be familiar with freely available geocoding, routing, and other geospatial services, and understand how to integrate them into interactive web maps.The exam also covers the Cesium.js virtual globe API, including its strengths, weaknesses, and practical examples of use. Students are expected to combine theoretical understanding with practical knowledge to evaluate and implement web mapping solutions effectively.

- **Introduction to Web Mapping APIs.** General concepts and purposes.
- **Open-Source Web Mapping APIs.** Overview of the most popular options, including pros and cons. Introducing Leaflet
- **Raster Map Data Formats.** TMS, WMS, and georeferenced image formats.
- **Raster Data Sources**. Typical providers and accessibility.
- **Vector Geodata Formats on Client Side**. GeoJSON, vector tiles, and their usage.
- **Geocoding Services.** Freely available options and integration into web maps.
- **Routing Services.** Overview of open routing services and their web integration.
- **Other Geospatial Services**. Additional services (e.g., POI search, spatial analysis) for web mapping.
- **Cesium.js Virtual Globe API**. Features, advantages, limitations, and example applications.

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- **Gábor Farkas: Mastering OpenLayers 3. https://www.packtpub.com/product/mastering-openlayers-3/9781785281006**
- **Paul Crickard: Leaflet.js Essentials. https://www.packtpub.com/product/leafletjs-essentials/9781783554812**
- **Gede Mátyás: The OpenLayers 5 API.** http://mercator.elte.hu/~saman/edu/ol5/
- Leaflet API reference. https://leafletjs.com/reference.html
- **CesiumJS tutorial and reference.** https://cesium.com/learn/cesiumjs-learn/

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Dr. Gede Mátyás, egyetemi docens, PhD**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree): ……..….